Week 5 Resources: Market Structure, Momentum, and When RL Can Win

Goal of Week 5

Weeks 3 and 4 taught you an uncomfortable but important lesson: on a pure random walk, a correctly working agent earns *nothing*, and near-zero P&L is the right answer. You now know what "RL working correctly" looks like on an unbeatable market.

Week 5 flips the question: if we put real structure *into* the market, does the agent find it? You will add one line to the price dynamics — an AR(1) return process — and everything else (state, actions, reward, evaluation) stays exactly as in Week 4. That makes this week a controlled experiment: any change in the result is caused by the one thing you changed.

By the end of this week you should be able to:

- Explain why no strategy can profit on a pure random walk.
- Define lag-1 autocorrelation and measure it from data.
- Describe an AR(1) process and what ρ controls (momentum vs. mean reversion).
- Train a PPO agent that clearly beats the random and buy-and-hold baselines — and explain *why* it can now.
- Reason about when a signal is worth trading, given a transaction cost.

Recap: Key Ideas from Week 4

Before moving on, confirm you can answer these:

- Why should a well-trained agent's average P&L be near zero on a random walk?
- What does the transaction cost term in the reward discourage?
- What do the three curves in your cumulative P&L plot represent, and what did they look like in Week 4?
- Why is `deterministic=True` used during evaluation?

If any of these feel shaky, re-read the Week 4 resources before continuing.

Why the Random Walk Was Unbeatable

In the Week 3–4 environment, each price return was drawn independently: knowing all past returns tells you *nothing* about the next one. Formally, the expected reward of any position is

$$\mathbb{E}\left[\,\text{position}_t \times r_{t+1}\,\right] = \text{position}_t \times \mathbb{E}[r_{t+1}] = \text{position}_t \times 0 = 0,$$

because the position is chosen *before* the return is drawn, and the return has mean zero no matter what happened before. Every strategy — random, buy-and-hold, or a billion-parameter neural network — has expected P&L of exactly zero. Add transaction costs and the expectation turns negative. This is the simplest model of an *efficient market*, and it is why your Week 4 curves all hugged the zero line.

The only way to earn positive expected reward is for $\mathbb{E}[r_{t+1} \mid \text{state}_t]$ to *depend on the state*. This week, it will.

Autocorrelation: Structure You Can Measure

The **lag-1 autocorrelation** of a return series is the correlation between each return and the one immediately after it:

$$\text{autocorr} = \text{corr}(r_t,\, r_{t+1}).$$

- **Positive** autocorrelation: an up move tends to be followed by another up move. Markets people call this *momentum* or *trending*.
- **Negative** autocorrelation: an up move tends to be followed by a down move. This is *mean reversion* — the price snaps back.
- **Zero** autocorrelation: the past says nothing about the future — the random walk you already know.

Autocorrelation is measurable from data with one line of NumPy (`np.corrcoef(r[:-1], r[1:])`), which means you can *verify* a market has structure before asking an agent to find it. Get into this habit now: next week, with real data, checking the data before blaming the algorithm becomes essential.

The AR(1) Process

The simplest process with tunable autocorrelation is the **autoregressive process of order 1**, or AR(1):

$$r_t = \rho\, r_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \qquad -1 < \rho < 1.$$

Each return is a fraction ρ of the previous return, plus fresh noise. Three facts worth knowing (no derivations needed):

- The lag-1 autocorrelation of an AR(1) series is exactly ρ. So the parameter you set is the statistic you should measure back — a built-in sanity check.
- The best prediction of the next return, given the current one, is $\mathbb{E}[r_{t+1} \mid r_t] = \rho\, r_t$. For $\rho > 0$, the next return probably has the *same sign* as the last one.
- $|\rho| < 1$ is required for the process to stay stable; at $|\rho| \geq 1$ returns grow without bound.

What the Optimal Policy Looks Like

Here is what makes this week special: for this market, *you can work out the good policy yourself*, like the hand-coded CartPole rule from Week 1.

Ignore costs for a moment. If the last return was r_t, the expected next return is ρr_t, so the expected one-step reward of holding position p is

$$\mathbb{E}[\text{reward}] = p \times \rho\, r_t \times 100.$$

For $\rho > 0$: if $r_t > 0$, go long; if $r_t < 0$, go short. In words: **follow the sign of the last return**. For $\rho < 0$ the rule flips: bet *against* the last move.

This gives you a rare luxury in RL: a known target behaviour. When you inspect your trained agent in Task 3, you are not just hoping the reward went up — you can check whether the policy *makes sense*.

The Edge-vs-Cost Trade-off

Transaction costs change the picture. Switching position costs cost $\times$ $|\Delta\text{position}|$, so a switch is only worth it when the expected gain from being on the right side, roughly $|\rho\, r_t| \times 100$ per step, outweighs the cost of getting there. Two consequences you should see in your Task 4 sweep:

- **Weak signal + high cost = don't trade.** When ρ is small and the cost is large, the best policy trades rarely or not at all, and P&L falls toward zero.
- **Small moves are not worth acting on.** Even with a strong signal, when $|r_t|$ is tiny the expected gain is tiny too, so a cost-aware agent should hold through small wiggles and only reposition after large moves.

This trade-off — *is the edge bigger than the cost?* — is arguably the central question of real trading, and you will now have felt it in a controlled setting.

A Word of Caution: Profit Changes Evaluation

In Weeks 3–4, "suspicious" meant an agent that claimed profit. Now profit is genuinely possible, which cuts both ways: a positive result can be real, or it can still be luck. Keep the Week 4 discipline:

- Evaluate over many fresh episodes (50, as in the harness), never one lucky run.
- Remember each training run is one random seed. If a result matters, re-run it and see how much it moves before believing it.
- Compare against the baselines every time. "The agent made money" means little; "the agent consistently beat random and buy-and-hold on fresh episodes" means something.

Connecting Back to the Five Questions

As always, ground the new ideas in the core RL framework:

- **Agent:** The PPO policy network, unchanged from Week 4.
- **Environment:** `TradingEnvV3` — same interface, but returns now follow an AR(1) process.
- **State:** $[r_{t-k+1}, \ldots, r_t, \text{current\_position}]$ — and note that for AR(1), the single most recent return is the informative part.
- **Actions:** Discrete — 0 = Go Long, 1 = Hold, 2 = Go Short.
- **Reward:** position $\times$ price_return $\times$ 100 $-$ cost $\times$ $|\Delta\text{position}|$ — unchanged.

One environment property changed, and suddenly the same agent, state, actions, and reward produce a completely different outcome. That is the deepest lesson of this week: **in applied RL, the environment — not the algorithm — usually decides what is possible.**

Suggested External Resources (Optional)

You only need this document and the assignment to complete Week 5. If you want more context:

- Any short introduction to autocorrelation and AR processes from a time-series course — you need only the intuition and the AR(1) definition.
- A blog-level explanation of momentum and mean reversion in markets (search “momentum vs mean reversion intuition”). Skip anything with heavy math.
- Stable-Baselines3 documentation — you are now using it on your third custom environment; skim the Custom Environments page once more and notice how little of it you still need.

Self-Check Before Moving On

You are ready for Week 6 (real market data) when you can:

- Explain, in two sentences, why no strategy beats a pure random walk.
- Measure the lag-1 autocorrelation of any return series with NumPy.
- State what ρ controls in an AR(1) process and what the optimal sign-following rule is for $\rho > 0$.
- Show a cumulative P&L plot where your trained agent clearly beats both baselines — and explain what changed since Week 4 to make that possible.
- Explain when a signal is *not* worth trading, using the words “edge” and “cost”.